

Human Interface Device-Driven Teleoperation of a Robot-Assisted Ureteroscopy Platform and its Digital Twin for Laser Lithotripsy

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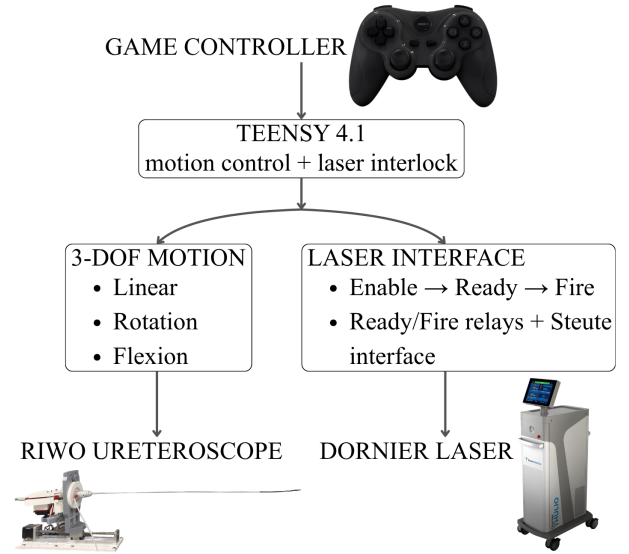
Abstract—We present a three-degree-of-freedom (3-DOF) robot-assisted ureteroscopy platform that integrates ureteroscope motion and laser lithotripsy control through a single human interface device (HID). A Wolf RIWO digital ureteroscope is actuated in linear translation, axial rotation, and distal-tip flexion using two stepper motors and one servo motor controlled through a Teensy 4.1 microcontroller. During teleoperation, joystick deflection proportionally commands motion speed in all three degrees of freedom, providing continuous proportional control. Position sensing supports axis monitoring, software limits, and homing, while the ST3020 flexion servo additionally incorporates internal magnetic position sensing and local closed-loop servo control. A Dornier MedTech lithotripsy laser is integrated through relay-based ready and fire interfaces adapted from the laser footswitch connection, with a software interlock requiring Enable and Ready states before Fire can be activated. Software-defined dead zones, axis gating, velocity limits, and homing functions further reduce unintended commands. In an initial benchtop demonstration, a novice operator under guidance successfully targeted and fired on eight stone targets using the integrated controller. The cumulative movement-to-fire time was 5 min 48 s, corresponding to a mean of 43.5 s per target, excluding laser firing duration and pauses between targets. This work demonstrates the feasibility of combining ureteroscope manipulation and laser control within a unified teleoperation interface.

Index Terms—Robot-Assisted Ureteroscopy, Surgical Robotics, Human Interface Device, Laser Lithotripsy, Teleoperation

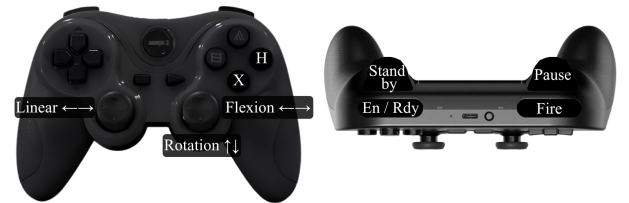
I. INTRODUCTION

Kidney stones are common, and ureteroscopy is an increasingly utilized treatment modality [1], [2]. Flexible ureteroscopy with laser lithotripsy requires the surgeon to coordinate ureteroscope insertion, rotation, distal-tip flexion, endoscopic visualization, and laser operation. These functions are normally distributed across manual ureteroscope controls and a separate laser footswitch.

Robotic flexible ureteroscopy has been investigated to improve instrument manipulation, ergonomics, teleoperation, and feedback-assisted control [3]–[8]. These systems demonstrate the potential of robotic interfaces to centralize ureteroscope manipulation and provide a foundation for computer-assisted navigation and training. The present work focuses specifically on integrating proportional three-degree-of-freedom ureteroscope motion and laser-state control within a single handheld HID, together with software-defined motion constraints and laser interlocks.



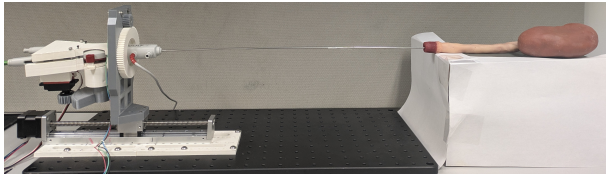
(a) Control architecture.



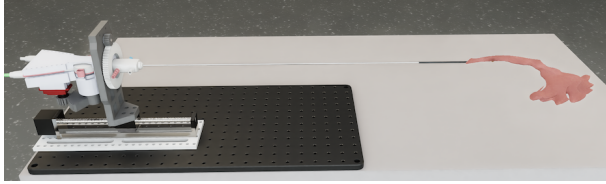
(b) Game controller mapping.

Fig. 1: HID-driven control framework. (a) Control architecture showing the game-controller input, Teensy 4.1 motion control and laser interlock, 3-DOF ureteroscope motion, position-feedback pathway, and laser interface. (b) Mapping of game-controller inputs to ureteroscope motion and laser command functions.

The objective of this work was to develop a human-operated robotic ureteroscopy system that places the primary motion and laser functions within a single handheld controller. The platform provides three controlled degrees of freedom—linear translation, axial rotation, and distal-tip flexion—together with integrated laser Enable, Ready, Fire, Standby, Pause, and Exit



(a) Robotic platform.



(b) Digital twin of robotic platform.

Fig. 2: Robotic platform and its digital twin for laser lithotripsy.

commands. The system was designed to provide proportional motion control, reject unintended joystick inputs, and prevent firing unless the required laser states have first been established.

This paper describes the mechanical and electronic platform, HID-based control architecture, flexion calibration, software control features, and an initial benchtop stone-targeting demonstration. The overall control architecture and HID command mapping are shown in Fig. 1.

II. SYSTEM DEVELOPMENT

The robotic platform and its digital twin for ureteroscopic laser lithotripsy are depicted in Fig. 2. These are built around a Wolf RIWO digital flexible ureteroscope [9]. Linear insertion and retraction are driven by a NEMA 14 stepper motor through a TB6600 driver, while axial rotation is driven by a NEMA 17 stepper motor through a second TB6600 driver. Both axes incorporate encoder feedback. During normal HID operation, joystick displacement is mapped proportionally to commanded velocity rather than to a fixed target position. Linear encoder data are used for position tracking, software travel limits, stall detection, and return to the 0-mm reference during HOME. Rotation encoder data are used for angular-position tracking, enforcement of the configured rotational range, and return to the 0° HOME reference. Distal-tip flexion is actuated by an ST3020 servo through a Waveshare bus-servo adapter. During HID operation, joystick displacement determines the rate at which the flexion target is advanced, providing speed-based operator control, while the servo maintains the commanded motor position through its internal closed-loop controller. The ST3020 incorporates an internal magnetic position sensor for this local position control. The complete experimental setup is shown in Fig. 3.

System control is centralized on a Teensy 4.1 programmed using the Arduino IDE. An Armor3 NuPlay wired game controller serves as the primary HID. Joystick displacement proportionally controls motion speed: one joystick commands

TABLE I: Motion Control Parameters and Software Limits

DOF	Max. Speed	Software Range	Limit Tolerance	HOME Tolerance
Linear	2.0 mm/s	0–177 mm	0.01 mm	0.50 mm
Rotation	30°/s	−180° to +180°	0.5°	1.0°
Flexion	35°/s	−270° to +270° tip	0.15–0.18° tip	1.0°

linear translation, while the second controls rotation and distal-tip flexion. Controller buttons provide HOME and laser-state commands. A separate controller-diagnostic implementation allows joystick axes and buttons from other compatible USB controllers to be identified and mapped through the serial interface.

A separate serial-command implementation was also developed for bench testing and diagnostics, allowing direct numerical motion commands and the same laser-state controls; however, the teleoperation experiment reported here used the HID interface exclusively.

The Dornier MedTech laser is connected through relay modules and a Steute breakout board adapted from the original laser footswitch interface. Separate signals control laser readiness and firing. The active-low Ready signal uses a 10-kΩ pull-up resistor, while the Fire signals use 10-kΩ pull-down resistors so that the control lines default toward inactive states. Laser firing is governed by a software-controlled state interlock. A Fire command is accepted only when both the Enable and Ready states are active. Attempts to fire before completing the sequence **Enable** → **Ready** → **Fire** are rejected by the controller, and the physical fire outputs are not activated. Standby, Pause, and Exit commands provide additional state control.

Joystick processing further limits unintended motion. A motion command is recognized only after joystick displacement exceeds approximately 12% from center. Linear motion uses an orthogonal-axis gate of 0.12, while rotation and flexion use an orthogonal-axis gate of 0.18. Excessive diagonal movement of the right joystick causes both rotation and flexion commands to be rejected rather than executed simultaneously. Maximum commanded velocities, software ranges, limit tolerances, and HOME thresholds are summarized in Table I.

Linear encoder scaling is approximately 0.004 mm per count, making the 0.01-mm software limit tolerance equivalent to approximately 2–3 encoder counts. Rotation encoder resolution corresponds to approximately 0.144° per count at the ureteroscope side, making the 0.5° tolerance approximately 3–4 encoder counts. Flexion is constrained to $\pm 270^\circ$ at the ureteroscope tip, corresponding to approximately $\pm 180^\circ$ of servo rotation. The ST3020 servo provides approximately 0.088° of motor rotation per step; because the calibrated flexion gain varies across the motion range, one servo step corresponds to approximately 0.15–0.18° of estimated distal-tip motion.

The HOME function returns linear translation, rotation, and flexion to their software reference states using the available position feedback. The values in Table I are software control parameters and should not be interpreted as independently measured physical positioning accuracy.

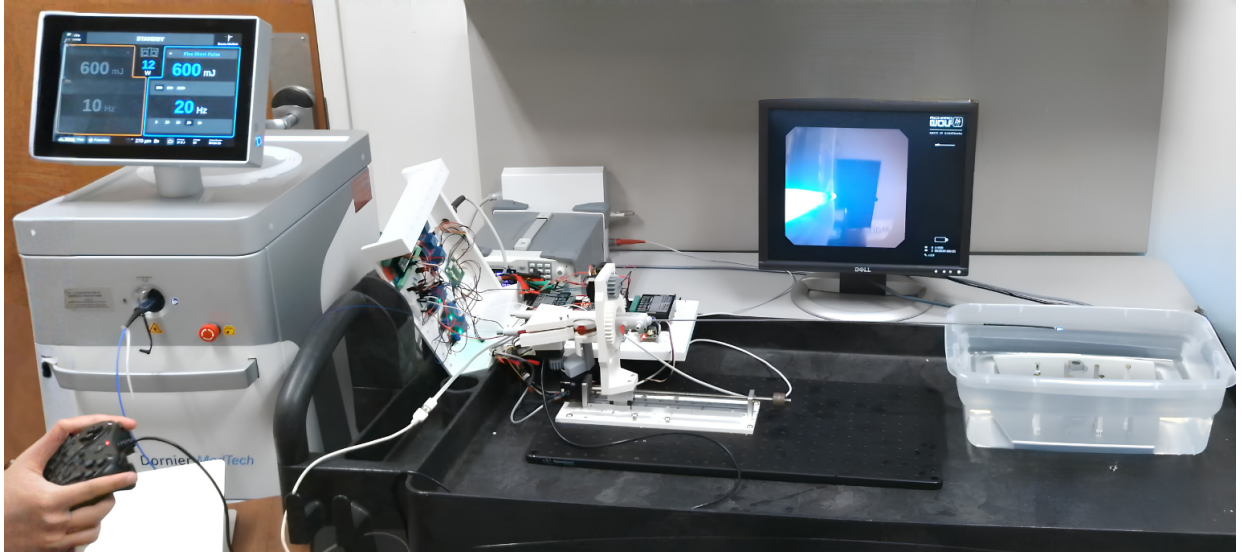


Fig. 3: Benchtop stone-targeting experiment showing the handheld controller, robot-assisted ureteroscopy platform, Dornier MedTech laser system, endoscopic display, and water-filled eight-target container.

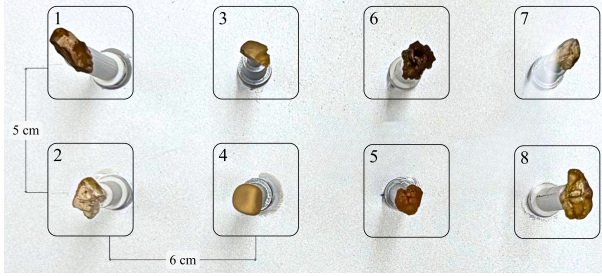


Fig. 4: Numbered arrangement of the eight stone targets used in the benchtop experiment. Numbers indicate the targeting sequence. Target spacing was 5 cm between rows and 6 cm between adjacent columns, measured center-to-center between the stone mounts.

III. BENCHTOP EVALUATION AND RESULTS

An initial benchtop experiment was performed to determine whether the complete system could be used to navigate among multiple stone targets and operate the lithotripsy laser entirely through the handheld controller.

Eight medium-sized training stone targets were arranged in a 2×4 pattern within the water-filled container, with 5 cm between rows and 6 cm between adjacent columns measured center-to-center between the stone mounts. Stone composition was not treated as an experimental variable in this feasibility demonstration. The ureteroscope distal tip and laser fiber remained submerged throughout the task.

The numbered targeting sequence and experimental geometry are shown in Fig. 4. A novice operator, guided by an instructor familiar with laser lithotripsy, was asked to navigate sequentially to all eight stones using only the handheld controller. The task did not require a stone to occupy a predetermined fraction of the endoscopic image. Instead, the

TABLE II: Stone Target Movement-to-Fire Times

Target	Movement Interval	Movement-to-Fire Time
1	0:00–0:46	46 s
2	0:52–1:21	29 s
3	1:27–2:05	38 s
4	2:09–2:41	32 s
5	2:46–4:15	89 s
6	4:23–4:50	27 s
7	5:15–6:21	66 s
8	6:27–6:48	21 s
Mean	–	43.5 s

operator visually aligned the laser with each stone in the ureteroscope-tip camera view and then fired on the target.

Endoscopic views at the time of laser firing for all eight targets are shown in Fig. 5. All eight targets were successfully acquired and fired upon. Movement-to-fire timing was measured from the start of movement toward each target until laser firing. Laser firing duration and pauses after firing before movement toward the next target were excluded. For clarity, the timing intervals are reported relative to the start of the first movement, defined as 0:00. Individual movement-to-fire times are summarized in Table II. The cumulative movement-to-fire time was 5 min 48 s, with a mean of 43.5 s per target.

The individual target times should not be interpreted as directly comparable measures of control performance because the stones occupied different positions within the 2×4 arrangement and therefore required different travel distances between targets. Target 5 required 89 s because the novice operator continued advancing and adjusting the ureteroscope until the stone occupied most of the endoscopic field of view, although this was not required by the task; the resulting close-up view is visible in Fig. 5. The 66-s maneuver toward target 7 followed a brief sticking event involving the laser fiber at

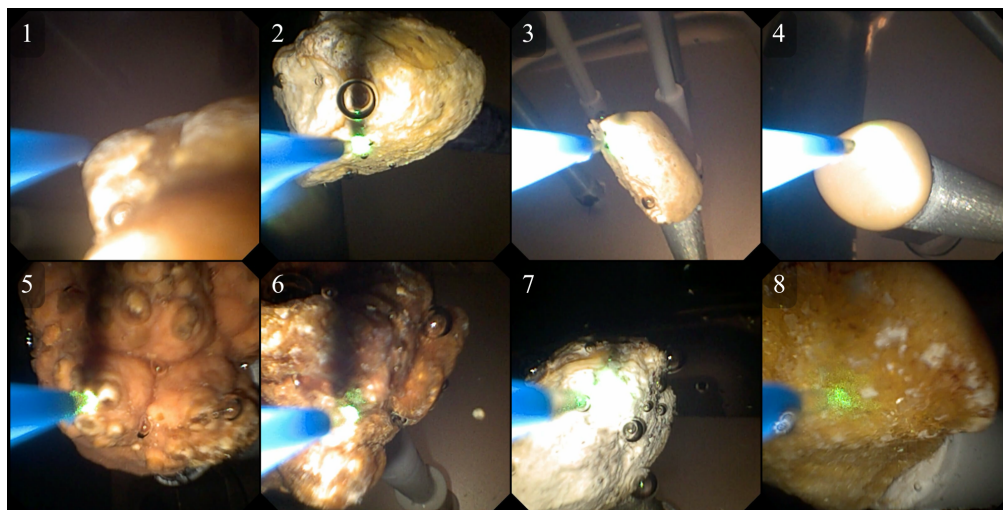


Fig. 5: Ureteroscopy-tip camera views at the time of laser firing for stone targets 1–8, shown in targeting order. The operator visually aligned the laser with each stone before firing.

target 6, which affected the subsequent handling.

This experiment was intended as a functional feasibility demonstration rather than a formal assessment of clinical accuracy, safety, or comparative usability. Nevertheless, it demonstrated that proportional translation, rotation, tip flexion, endoscopic targeting, and laser activation could be coordinated through a single HID and that a novice operator could complete the prescribed task after instruction.

IV. CONCLUSION AND FUTURE WORK

We developed a 3-DOF robot-assisted ureteroscopy platform that combines proportional ureteroscopy motion and laser lithotripsy operation within a unified handheld interface. Normal teleoperation is velocity based across all three degrees of freedom, while position sensing supports axis monitoring, software limits, and coordinated HOME operation. The ST3020 flexion servo additionally provides local closed-loop position control through its internal magnetic position sensor. The system also incorporates empirically calibrated tip flexion, software-defined motion constraints, and a state-dependent software interlock for laser firing. An initial eight-target benchtop experiment demonstrated operation of the complete navigation and firing workflow through the controller.

Future work will focus on quantitative measurement of positioning accuracy and repeatability, validation with multiple operators, and evaluation in anatomically representative ureteroscopy environments. We are also developing an NVIDIA Isaac Sim ureteroscopy environment with Meta Quest 3 control to connect virtual and physical ureteroscopy operation for training and system development. Future versions will further develop computer-assisted targeting and supervised autonomy while retaining surgeon control.

The prototype hardware will also be transitioned from development wiring toward a more robust implementation using custom printed circuit boards, locking connectors, enclosed electronics, strain relief, and electrically isolated interfaces.

These steps, together with formal motion and safety testing, will be required before evaluation in clinically representative settings.

Together, these developments establish a pathway from the current teleoperated prototype toward a robust platform for surgical training, computer-assisted targeting, and supervised robotic ureteroscopy.

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